

Ayaan Rahman – BG1/R1

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About Me

Member of Knights of Peter Claver and a Software Engineering candidate focused on building scalable backend systems and clean, reliable APIs. Strong in shipping production-ready features end-to-end: designing, implementing, testing, deploying, and monitoring. I optimize for maintainability, latency, and measurable outcomes (automation, cost reduction, throughput, reliability).

Education

University of California, Berkeley

Bachelor of Science in Computer Science

2022 – 2026

Experience

Software Engineer Intern

Stripe — San Francisco, CA

May 2025 – Aug 2025

- Developed high-throughput REST APIs in Java (Spring Boot) serving 120K+ daily requests with 99.98% uptime.
- Reduced P95 API latency by 37% by redesigning the database indexing strategy and eliminating N+1 query patterns.
- Implemented an idempotent webhook processing system using Redis + background workers, reducing duplicate transaction errors by 62%.

Backend Engineer (Contract)

Fin Edge Analytics — Remote

Jan 2024 – Apr 2025

- Designed microservices in Python (FastAPI) with PostgreSQL handling 50K+ monthly user interactions.
- Built an automated ETL pipeline that processed 3M+ financial records weekly, reducing manual reconciliation time by 58%.
- Optimized SQL joins, query execution plans, improving reporting dashboard load time by 41%.

Projects

SwiftScale (E-Commerce Engine)

Tech Stack: React, Node.js, Express, PostgreSQL, Redis, Stripe API, Docker.

- **Optimized database performance by 40%** through the implementation of Redis caching and PostgreSQL indexing, reducing average API response times from 350ms to **210ms**.
- **Integrated Stripe API** with robust webhook handling, achieving a **99.9% successful transaction rate** and automating post-purchase email triggers for 5,000+ mock users.
- **Reduced frontend bundle size by 25%** using code-splitting and Lazy Loading in React, resulting in a **1.5s improvement** in Largest Contentful Paint (LCP).

Sentinel (Real-Time Monitoring Dashboard)

Tech Stack: Next.js, TypeScript, Go (Golang), MongoDB, Prometheus, Socket.io, AWS.

- **Improved incident response time by 60%** by architecting a real-time notification system using WebSockets (Socket.io) that alerts admins within **<500ms** of a service failure.
- **Developed a custom data-polling engine** that processed over **10,000 data points per minute** with a CPU utilization rate of **less than 15%** on a standard t2.micro instance.
- **Automated deployment workflows** using GitHub Actions, reducing manual deployment errors by **90%** and cutting release cycle time from 1 hour to **6 minutes**.

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL

Frameworks: Spring Boot, FastAPI, Node.js, React

Databases: PostgreSQL, MySQL, Redis

Cloud & DevOps: Docker, AWS (EC2, S3, RDS), GitHub Actions, Linux

Core Strengths: REST API Design, Microservices, System Design, Query Optimization, Concurrency, Testing